

Trinh Dang Khoa

Alex Trinh

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Availability: September 7 – December 14, 2026 (3+ months)

ABOUT ME

Methodical engineer who thrives in team-based environments, contributing reliable work while actively learning from peers. Adapts quickly to new MCU platforms and eager to grow within automotive embedded software development.

SKILLS

Technical Skills

- **Hardware & MCUs: NXP S32K144** (Automotive), **STM32** (Bluepill, Nucleo), MAX17048, RC522.
- **Programming:** C/C++ (C++11/C++14), Embedded Firmware, Bare-metal Development, OOP, Generic Programming (Templates).
- **Automotive:** AUTOSAR (BSW, RTE fundamentals), CAN, LIN, I2C, SPI, UART.
- **Tools:** S32 Design Studio, STM32CubeIDE, MAAZ EDU (FPT Software), Oscilloscopes, Hardware Debugging.

Soft Skills

- **Teamwork & Communication:** Effective collaborator in team-based projects, clear technical communicator.
- **Analytical Thinking:** Strong problem-solving skills applied to hardware debugging and firmware troubleshooting.
- **English Proficiency: IELTS 7.0** — capable of working effectively in international engineering teams.
- **Adaptability:** Fast learner across multiple MCU platforms (NXP, STM32) and development environments.

PROJECTS & EXPERIENCE

Automotive ECU Firmware & Diagnostics (NXP S32K144) 2025
Hardware Debugging Project

Developed **bare-metal firmware** for the **NXP S32K144** automotive microcontroller, implementing a dual-board **ECU** communication system using **I2C**. Performed **integration testing** and physical layer verification using an **oscilloscope** to analyze and debug communication signals, ensuring reliable hardware interfacing. Documented test results and debugging procedures.

Embedded Battery Monitoring System (STM32) 2025
Firmware & Embedded Systems Project

Engineered **C/C++ firmware** for an **STM32 Bluepill** microcontroller interfaced with a **MAX17048** fuel gauge IC via **I2C**. Implemented a **Coulomb counting** algorithm to provide real-time, accurate monitoring of capacity and power consumption for custom-built 4S3P LiFePO4 battery packs.

RF Hardware Interfacing & Protocol Analysis 2024
Independent Research

Programmed an **RC522 RFID** module to interface with **MIFARE Classic** cards, executing low-level memory operations and **UID** modifications. Conducted hands-on analysis of **RFID communication protocols**, demonstrating a strong understanding of physical layer hardware interfacing and security considerations.

EDUCATION

FPT University 2024–Present
Automotive Software Engineering, Bachelor of Information Technology

3rd year student

FPT Academy 2025–Present
Automotive Software Development

Training program